Build your SeedPeers micro greenhouse

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A micro-greenhouse made from a recycled 2 L PET bottle of Coca-Cola. It has a grow light and an ESP32 that you control from your smartphone.



Figure 1: The finished micro-greenhouse

What you need

Materials

#	Item	Qty	Notes
1	Empty 2 L PET bottle with cap	1	Clear plastic, label removed
2	Plywood board	1	About 1cm thick. You cut a disc from it the size of the bottle
3	LED ring grow light	1	Outer diameter ___ mm (TODO), ___ V (TODO) A possible choice
4	ESP32 dev board (ESP32-WROOM-32)	1	Mounted on a perfboard
5	Dual MOSFET driver module	1	Switches/dims the LED ring. A possible choice

#	Item	Qty	Notes
6	Perfboard + jumper wires	1	
7	USB power supply + cable	1	5 V, > ___ A (TODO)
8	Small wood screws	4–6	To fix the board and the bottle top to the disc
9	Hot-glue sticks	–	
10	Small plant pot + soil + seedling	1	Must fit inside the bottle

Tools

- Cutter / sharp scissors
- Jigsaw (or coping saw)
- Hot-glue gun
- Screwdriver
- Pencil, masking tape
- Soldering iron *(optional, for the perfboard)*

!!! Danger wear eye protection when you use the jigsaw. Keep fingers away from the blade and be careful with the cutter on curved plastic. Hot glue causes burns.

Step 1 – Cut the bottle



Figure 2: The bottle cut in two parts

1. Wrap a strip of masking tape around the bottle at ___ **cm** from the base (TODO). Use it as your cutting guide.
2. Cut along the tape with the cutter, then finish the cut with scissors.
3. You now have two parts:

- the **bottom part**, which holds the plant,
- the **top part** (shoulder + cap), which becomes the lid.

Tip: smooth the cut edges so they don't scratch.

Step 2 – Check the fit (basic greenhouse)



Figure 3: The basic greenhouse without electronics

Put the potted seedling in the bottom part and slide the top part over it. This is already a working passive greenhouse. The next steps add the light and the electronics.

Step 3 – Cut the wooden disc



Figure 4: Tracing and cutting the disc with a jigsaw

1. Place the top part of the bottle on the plywood and trace around it with a pencil.
2. Cut out the disc with the jigsaw. Stay slightly **inside** the line so the disc fits snugly inside the bottle.
3. Sand the edges.

Step 4 – Mount the LED ring

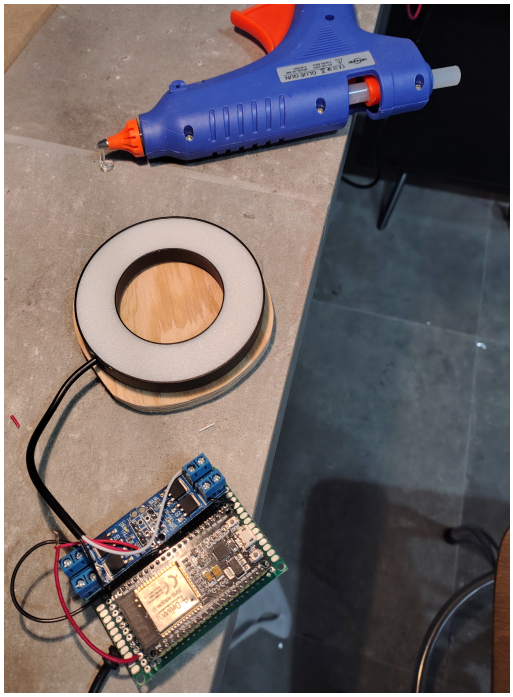


Figure 5: LED ring glued to the disc, next to the electronics

1. Hot-glue the LED ring to the **underside** of the disc so the light points down at the plant.
2. Route the LED cable to the top side of the disc.

Step 5 – Mount the electronics

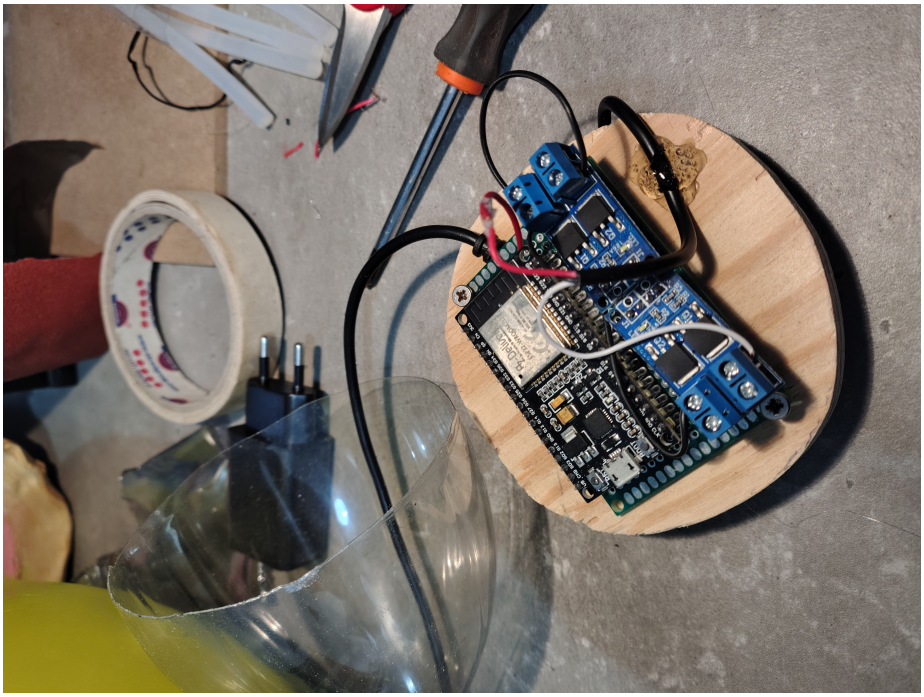


Figure 6: ESP32 and MOSFET module screwed to the disc

1. Screw the perfboard with the ESP32 and the MOSFET module onto the **top side** of the disc.
2. Wire the parts as follows:

From	To
ESP32 GPIO ___ (TODO)	MOSFET module IN1
ESP32 GND	MOSFET module GND
LED ring + / –	MOSFET module OUT1
USB 5 V / GND	ESP32 VIN / GND and MOSFET module V+

3. Fix the cables with a drop of hot glue (strain relief).

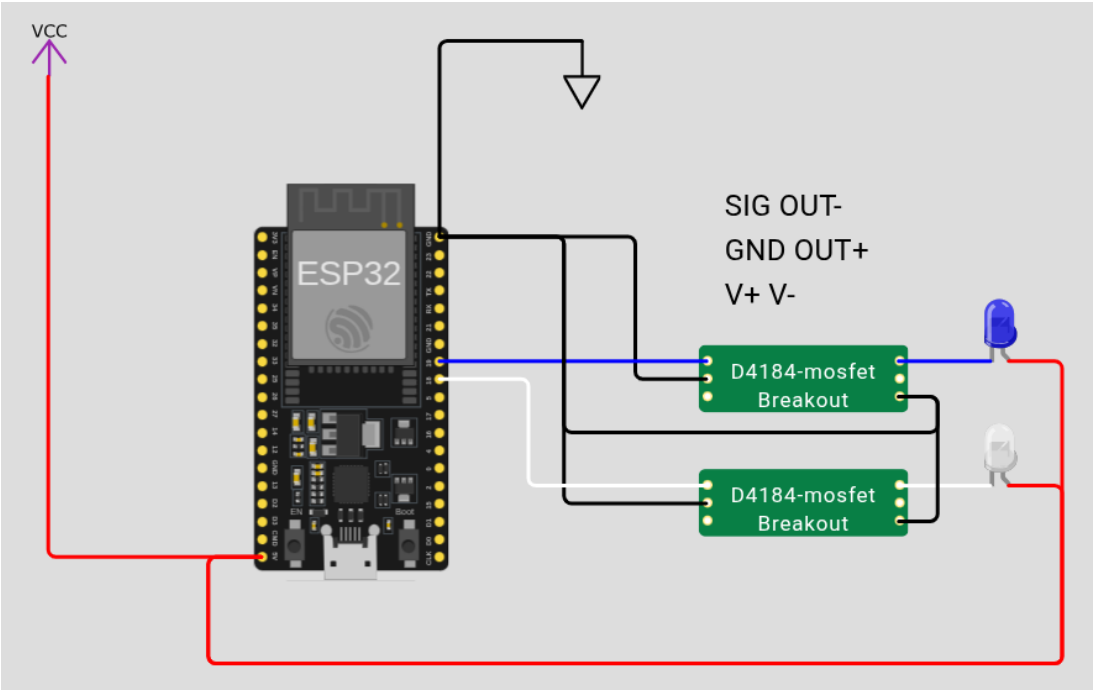


Figure 7: alt text

```
const int PIN_Q1_SPECTRUM_1 = 19; // PWM Canale 1 (Rosso/Blu)
const int PIN_Q2_SPECTRUM_2 = 18; // PWM Canale 1 (Rosso/Blu)
const int PWM_FREQ = 5000;
const int PWM_RESOLUTION = 8;

void setLampState(int brightnessQ1, int brightnessQ2) {
  // Con il Core v3.x usi direttamente il numero del PIN
  ledcWrite(PIN_Q1_SPECTRUM_1, brightnessQ1);
  ledcWrite(PIN_Q2_SPECTRUM_2, brightnessQ2);
}

void setup() {
  Serial.begin(115200);

  // Sintassi ESP32 Core v3.x: associa pin, frequenza e risoluzione in un solo passaggio
  ledcAttach(PIN_Q1_SPECTRUM_1, PWM_FREQ, PWM_RESOLUTION);
  ledcAttach(PIN_Q2_SPECTRUM_2, PWM_FREQ, PWM_RESOLUTION);

  setLampState(0,255);
  Serial.println("ready");
}

void loop() {
  Serial.println("ramping up");
  for (int duty = 0; duty <= 255; duty += 5) {
    setLampState(duty, 255-duty);
    Serial.print("duty: ");
    Serial.println(duty);
    delay(20);
  }

  delay(1000);
```

```
  Serial.println("ramping down");
  for (int duty = 255; duty >= 0; duty -= 5) {
    setLampState(duty, 255-duty);
    Serial.print("duty: ");
    Serial.println(duty);
    delay(20);
  }
}
```

Step 6 – Close the lid



Figure 8: Top part of the bottle screwed onto the disc

1. Pass the power cable through the bottle neck. Drill a hole in the cap or leave the cap off.
2. Slide the top part of the bottle over the disc and fix it with small screws through the plastic into the edge of the disc.

Step 7 – Assemble the greenhouse



Figure 9: The fully assembled greenhouse

1. Put the potted plant in the bottom part.
2. Rest the lid unit (disc + electronics + top part) on the bottom part.
3. Plug the USB cable into the power supply.

Step 8 – Upload the firmware

1. Install the Arduino IDE (or PlatformIO) and the ESP32 board support.
2. Open the firmware: `TODO: link to repository / sketch`.
3. Set your parameters (TODO: Wi-Fi / Bluetooth, RemoteXY token, …).
4. Upload the sketch to the ESP32 over USB.

Step 9 – Turn on the grow light



Figure 10: Grow light on

When powered, the LED ring lights up the plant with the grow spectrum (pink/purple light).

Step 10 – Control it from your smartphone

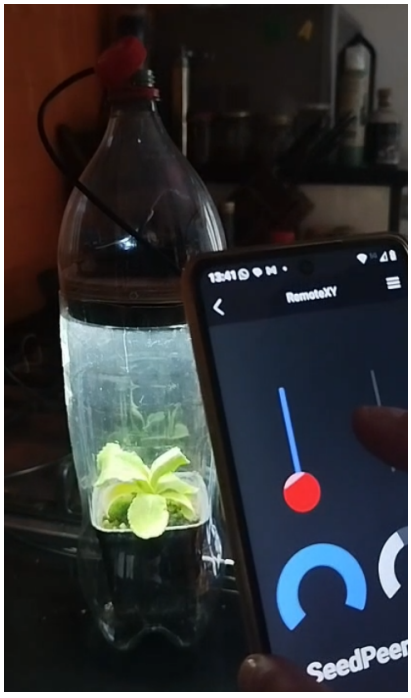


Figure 11: Control from the RemoteXY app

1. Install **RemoteXY** on your phone (Android / iOS).
2. Connect to the **SeedPeers** device (TODO: connection method).
3. Use the controls to set the light intensity and TODO (other controls).
